

DATA SCIENCE – STUDY MATERIAL

Exam-oriented notes covering fundamental concepts, model development, regression, evaluation, and applications of Data Science.

1. Introduction to Data Science

Definition

Data Science is an interdisciplinary field that uses statistics, mathematics, programming, machine learning, and domain knowledge to extract useful information and insights from data.

Key objectives

The main objectives are to collect data, clean and transform it, discover patterns, build predictive models, evaluate results, and support better decision-making.

Importance

Data Science helps organizations understand past events, predict future outcomes, automate decisions, identify trends, and solve complex problems.

2. Scope of a Data Science Project

Major stages

A Data Science project commonly includes problem definition, data collection, data understanding, data preprocessing, exploratory data analysis, feature engineering, model building, model evaluation, deployment, and monitoring.

Project scope

The scope defines the problem to be solved, required data, expected output, constraints, evaluation criteria, and intended users of the solution.

3. Data Scientist

Definition

A data scientist is a professional who analyzes data and develops models or analytical solutions to solve real-world problems.

Skills

Important skills include Python or R programming, SQL, statistics, machine learning, data visualization, communication, and domain knowledge.

4. Data Sources

Primary data source

A primary data source is data collected directly by the researcher or organization for a specific purpose. Examples include surveys, interviews, experiments, observations, and questionnaires.

Secondary data source

Secondary data is data collected previously by another person or organization. Examples include government datasets, published reports, databases, and research repositories.

5. Data Preprocessing

Definition

Data preprocessing is the process of preparing raw data for analysis or machine learning.

Common steps

Common steps include handling missing values, removing duplicates, correcting inconsistent data, detecting outliers, encoding categorical variables, scaling numerical features, and selecting useful features.

Importance

Good preprocessing improves data quality and can make machine learning models more reliable and effective.

6. Exploratory Data Analysis (EDA)

Definition

Exploratory Data Analysis is the process of examining and summarizing data to discover patterns, relationships, distributions, anomalies, and useful insights.

Methods

EDA uses summary statistics, frequency tables, histograms, box plots, scatter plots, correlation analysis, and other visualizations.

7. Mean, Median and Mode

Mean

Mean is the arithmetic average. It is calculated by adding all observations and dividing by the number of observations.

Median

Median is the middle value after arranging observations in ascending or descending order. For an even number of observations, it is the average of the two middle values.

Mode

Mode is the value that occurs most frequently in a dataset.

8. Regression

Definition

Regression is a supervised learning technique used to predict a continuous numerical value from one or more input variables.

Applications

Examples include predicting house prices, sales, temperature, student scores, and demand.

Basic idea

A regression model learns the relationship between independent variables and a dependent variable using training data.

9. Polynomial Regression

Definition

Polynomial Regression models a relationship between variables using polynomial terms such as x^2 , x^3 , and higher powers.

Use

It is useful when the relationship between input and output is curved rather than approximately linear.

Caution

Very high polynomial degrees can make a model overly complex and lead to overfitting.

10. Ridge Regression

Definition

Ridge Regression is a regularized form of linear regression that adds an L2 penalty to the model's loss function.

Purpose

It helps reduce overfitting and controls the size of model coefficients, especially when predictors are highly correlated.

Regularization

The regularization strength is controlled by a parameter commonly represented by alpha or lambda.

11. Overfitting and Underfitting

Overfitting

Overfitting occurs when a model learns the training data too closely, including noise, and performs poorly on unseen data.

Underfitting

Underfitting occurs when a model is too simple to capture important patterns in the data and performs poorly even on training data.

Balance

A good model should learn meaningful patterns while maintaining good performance on unseen data.

12. Generalization Error

Definition

Generalization error is the difference between a model's expected performance on unseen data and its predictions on that unseen population.

Importance

A model with low generalization error usually performs well on new data, while a model with high generalization error may have overfitted or failed to learn useful patterns.

13. Hyperparameters

Definition

Hyperparameters are settings chosen before or during model training that control how a machine learning algorithm learns.

Examples

Examples include learning rate, number of trees, tree depth, regularization strength, number of neighbors, and batch size.

Tuning

Hyperparameter tuning searches for suitable values using methods such as grid search, random search, or other optimization strategies.

14. Model Evaluation

Purpose

Model evaluation measures how well a trained model performs using suitable metrics and validation data.

Regression metrics

Common regression metrics include Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared (R^2).

Classification metrics

Common classification metrics include accuracy, precision, recall, F1-score, and ROC-AUC.

Validation

A dataset may be divided into training, validation, and test sets so that model selection and final performance can be evaluated fairly.

15. Pipeline in Model Development

Definition

A machine learning pipeline is an organized sequence of data processing and modeling steps executed in a consistent order.

Typical flow

A typical pipeline is: data collection → preprocessing → feature transformation → model training → validation → evaluation → deployment.

Benefits

Pipelines improve reproducibility, reduce manual errors, keep preprocessing consistent, and simplify model development and deployment.

16. Complete Data Science Project Life Cycle

Step 1 – Problem definition

Clearly identify the business or real-world problem and define measurable objectives.

Step 2 – Data collection

Collect relevant data from primary or secondary sources.

Step 3 – Data understanding

Inspect variables, data types, distributions, quality, and relationships.

Step 4 – Data preprocessing

Clean, transform, encode, scale, and prepare the data.

Step 5 – EDA

Explore patterns, relationships, trends, and anomalies.

Step 6 – Feature engineering

Create, transform, or select useful features.

Step 7 – Model development

Choose suitable algorithms and train models.

Step 8 – Model evaluation

Compare models using appropriate evaluation metrics.

Step 9 – Deployment

Make the selected model available for practical use.

Step 10 – Monitoring

Monitor performance and update the model when data or requirements change.

17. Applications of Data Science

Healthcare

Disease risk analysis, medical image analysis, patient monitoring, and resource planning.

Finance

Fraud detection, risk assessment, credit analysis, and forecasting.

Education

Student performance analysis, personalized learning, and dropout prediction.

Retail

Recommendation systems, customer analysis, demand forecasting, and inventory planning.

Transportation

Route optimization, traffic analysis, demand prediction, and predictive maintenance.

Agriculture

Crop monitoring, yield prediction, soil analysis, and precision farming.

18. Important 2-Mark Questions

1.

- Define Data Science.

2.

- What are the different scopes of a Data Science project?

3.

- What is a data scientist?

4.

- What is a primary data source?

5.

- Define data preprocessing.

6.

- Define Exploratory Data Analysis.

7.

- Define mean.

8.

- What is a hyperparameter?

9.

- Define regression.

10.

- Differentiate between overfitting and underfitting.

End of Study Material

Use this material for revision, AI-generated summaries, flashcards, and quiz generation.